آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre SSC-II General Mathematics Annual Examinations 2024

Introduction

This document has been prepared for the teachers and candidates of Secondary School Certificate (SSC) Part II (Class X) General Mathematics. It contains comments on candidates' responses to the 2024 SSC-II Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

Candidates performed reasonably well in some concepts, such as, Mode, Square Root by long division, Linear Equations and Inequalities. However, candidates who did not score well mostly failed to understand the demands of the questions, often misinterpreting the command words and the stimuli.

Mentioned below are few concepts that teachers need to focus so that the candidates may perform better.

- Simple Interest
- HCF and LCM by Division Method
- Quadratic Equation
- Fundamentals of Geometry
- Area and Volume

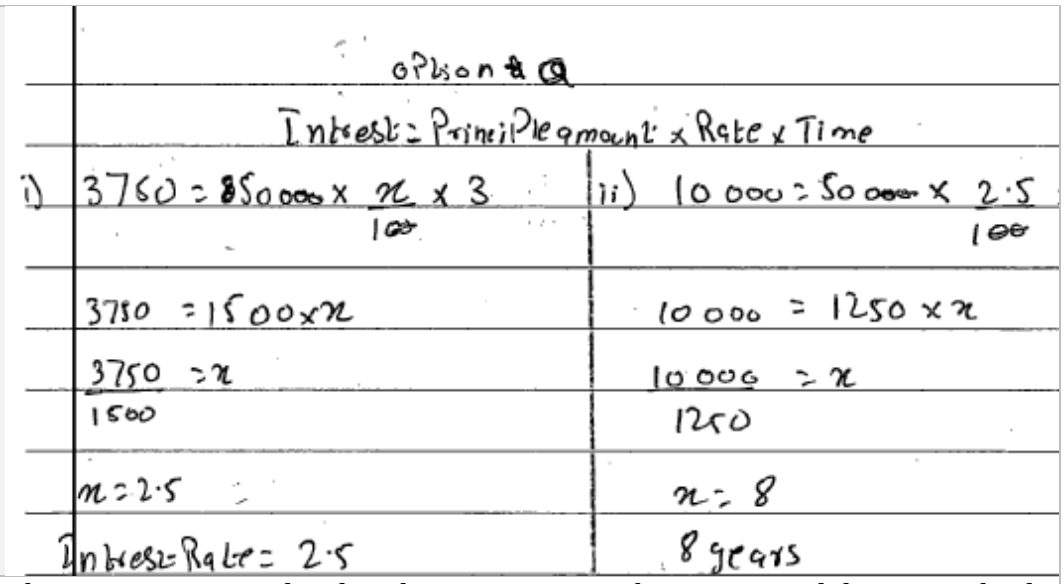
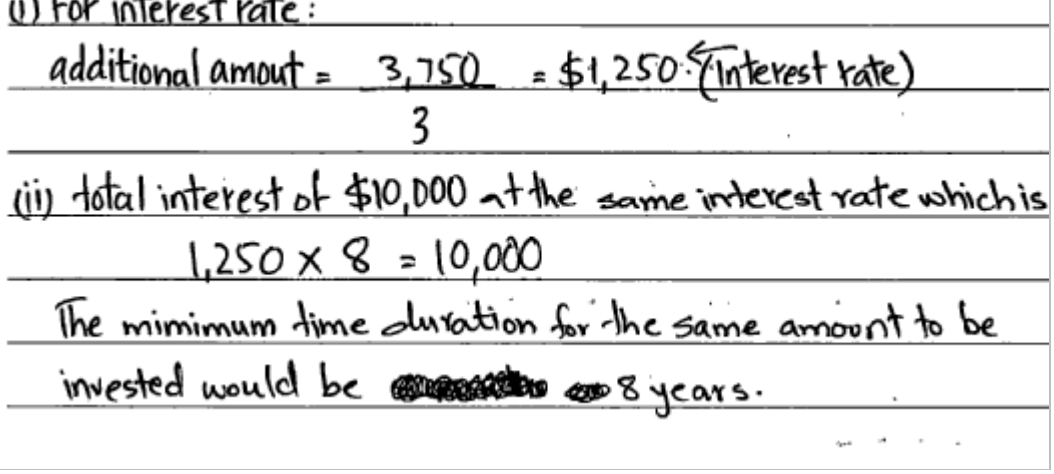
Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS
Constructed Response Questions (CRQs)

Question No. 1a

Candidates were given the choice to attempt any ONE out of the two questions: 1a and 1b. Most of the candidates attempted 1a.

Question Text	An investor has deposited $50,000 for a certain period of time. i. If he received an additional amount of $3,750 at the end of 3 years, then determine the interest rate earned by the investor. ii. If the investor aims to receive a total interest of $10,000 at the same interest rate, then what is the minimum time duration that the same amount should be invested?
SLO No.	11.3.2
SLO Text	Solve problems related to simple interest.
Max Marks	3
Cognitive Level	A*
Checking Hints	i. 1 mark for substituting the values correctly. 1 mark for calculating the rate. ii. 1 mark for calculating number of years.
Overall Performance	This simple interest question seemed straightforward, but many candidates misunderstood it, leading to mistakes in either the first or the second part of the problem.
Description of Better Responses	Better responses demonstrated that candidates effectively followed the correct procedure to solve the question. They began by selecting the appropriate formula, then input the provided values of principal amount and interest to calculate the interest rate and number of years accurately.

Image of Better Response	
Description of Weaker Responses	The common error related to this question was that many candidates mistakenly treated an additional amount as interest. Some candidates also mistakenly subtracted it from the principal amount instead of adding it, causing errors in their calculations.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

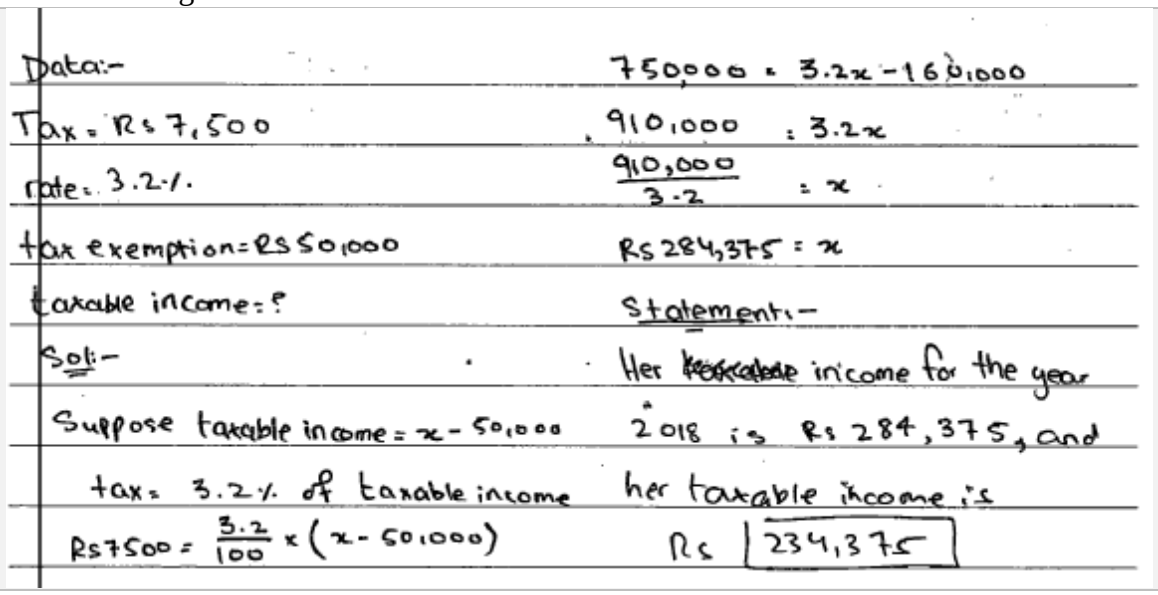
Maximising SLO Achievement	**Preferred Pedagogy** Used for this SLO**	**Assessment Strategies**
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

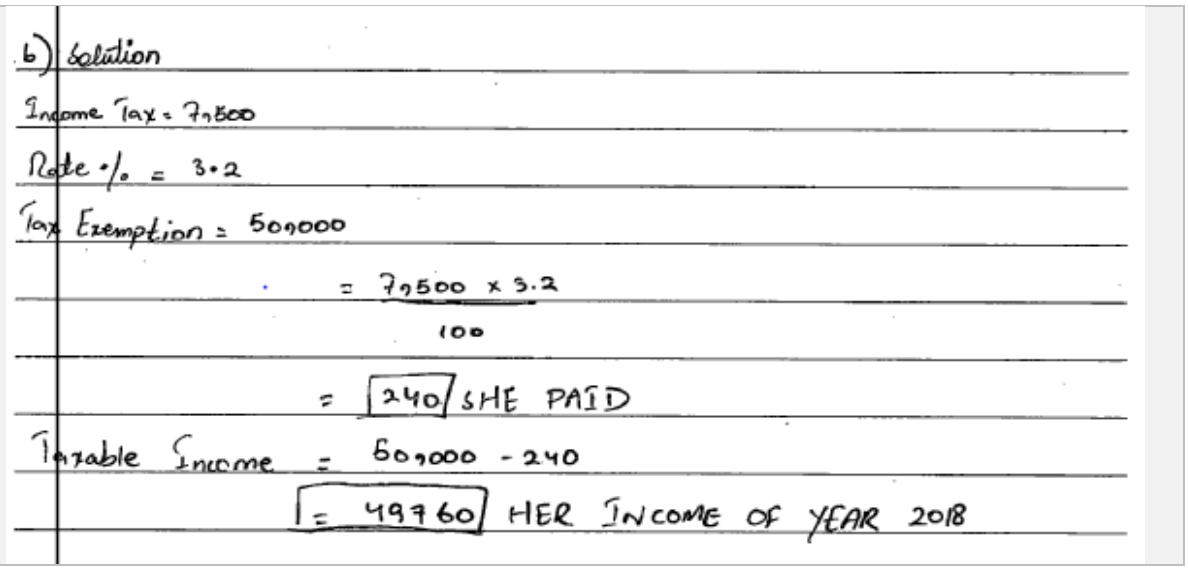
	** For description of each Pedagogy, refer to Annexure A	

Any Additional Suggestion:

1. Provide students with simple case studies involving real businesses dealing with simple interest calculations.
2. Have students analyse current news articles related to interest rates and financial markets.

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

Question No. 1b	
Question Text	In the year 2018, Halimah paid an income tax of Rs 7,500 at the rate of 3.2% while she was eligible for a tax exemption of Rs 50,000. Find out her taxable income for the year 2018.
SLO No.	12.3.3
SLO Text	Calculate net income taking into account assorted deductions. (income tax, etc).
Max Marks	3
Cognitive Level	A
Checking Hints	1 mark for making correct expression for taxable income 1 mark for substituting correctly in income tax amount formula 1 mark for total income
Overall Performance	Overall, the candidates' performance was below expectations. Those who grasped the distinction between taxable income and exempted amounts performed well, demonstrating a clear understanding of the question.
Description of Better Responses	Better responses showed that candidates were able to identify the important information from the question correctly by using right concepts and right formula. They were also able to write accurate expression for finding taxable income and then calculating the exact amount through calculations.
Images of Better Responses	
Description of Weaker Responses	Weaker responses lost marks by mistakenly treating the given amount as total income and calculating tax on the exempted amount. Others calculated tax on the given amount itself, indicating a lack of conceptual understanding or inability to understand the question's requirements.

Images of Weaker Responses	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Preferred Pedagogy Used for this SLO**	**Assessment Strategies**
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Any Additional Suggestion:		

Question No. 2	
Question Text	A group of students conducted a survey to find out the number of hours they spent studying per day. The results are given in the following table. Number of hours: 0 – 2 | 2 – 4 | 4 – 6 | 6 – 8 | 8 – 10 Number of Students: 4 | 7 | 12 | 7 | 1 Estimate the mode of the data using graphical methods.

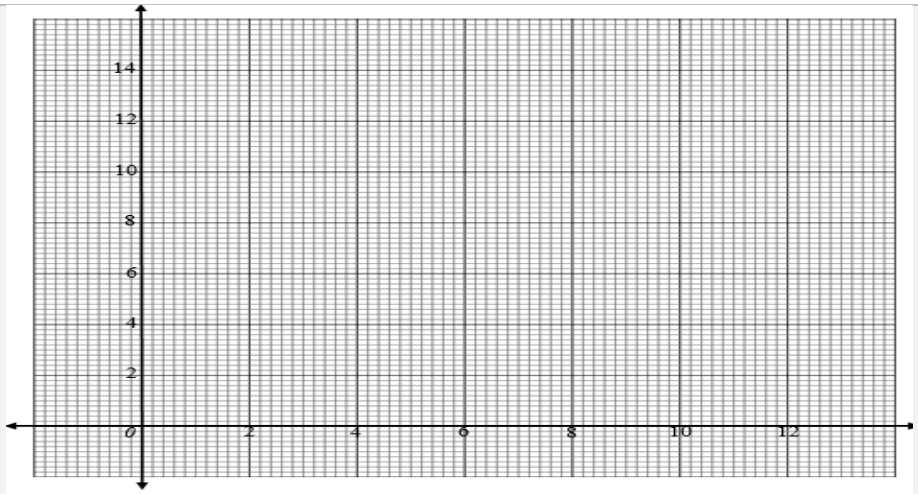
	
SLO No.	13.3.2
SLO Text	Estimate the median and mode graphically.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for writing correct axis 1 mark for making histogram 1 mark for taking intersections from the corners of the highest class 1 mark for finding correct mode
Overall Performance	This question on histograms required a solid grasp of basic graphical skills , particularly the concept of mode. Overall, candidates performed slightly above average on this question.
Description of Better Responses	Candidates who earned full marks demonstrated a clear understanding of constructing histograms. They accurately marked class boundaries on the x-axis and frequencies on the y-axis. Moreover, these candidates showed conceptual understanding in identifying the mode graphically.

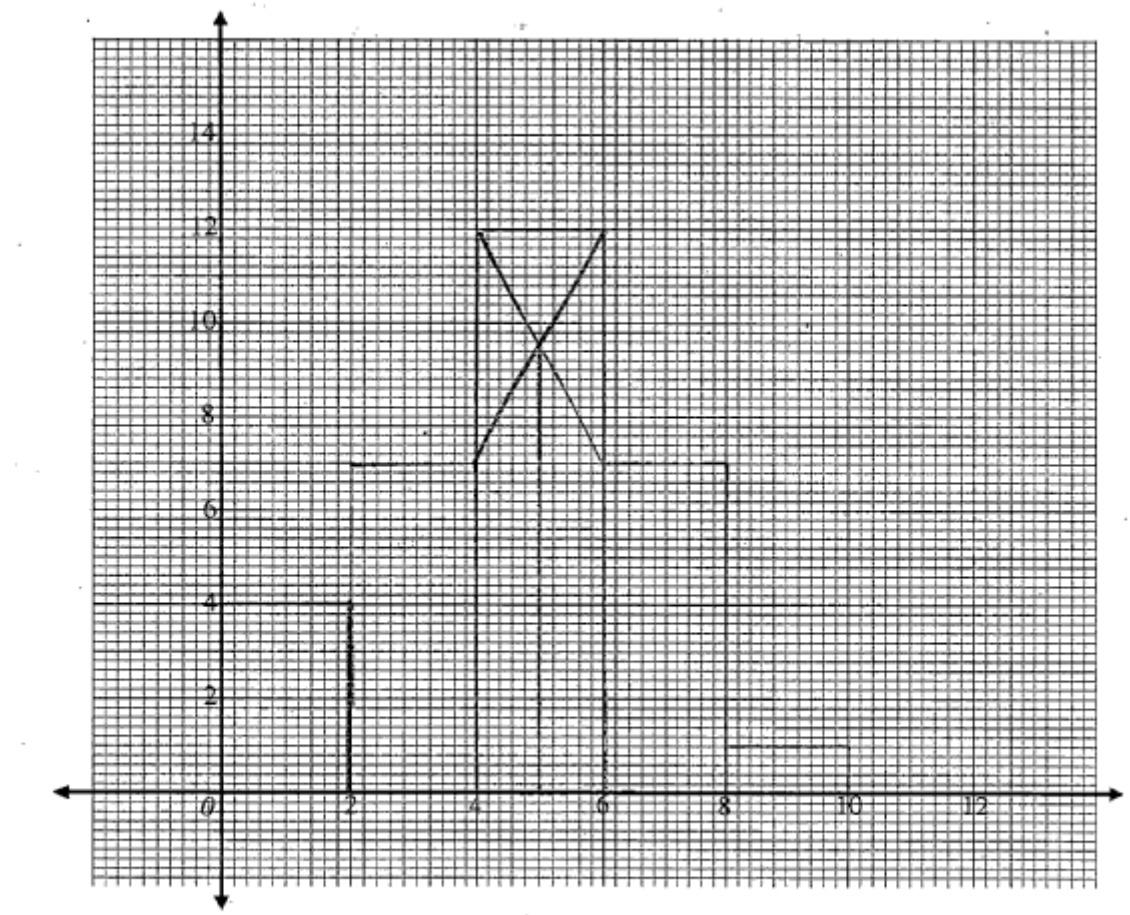
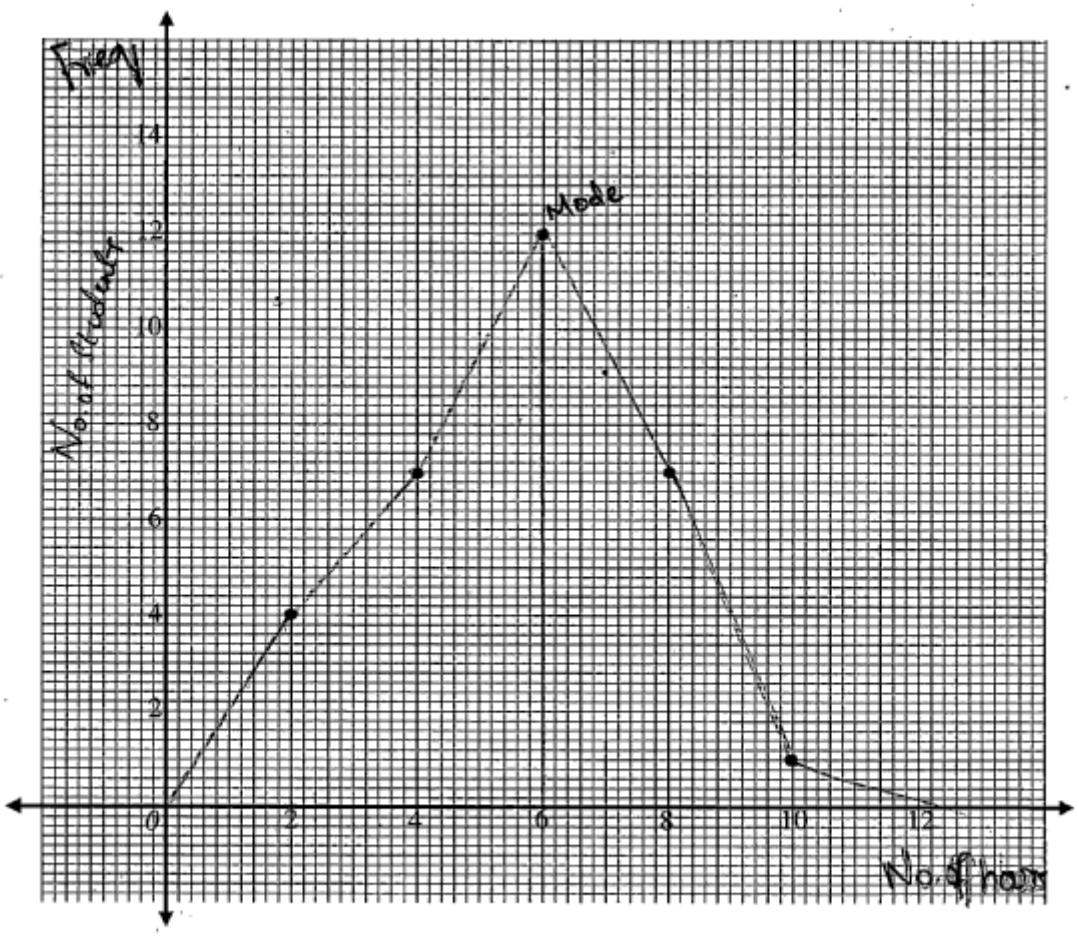
Image of Better Response	Estimate the mode of the data using graphical methods.  The Mode of the data is 5.
Description of Weaker Responses	Weaker responses struggled with identifying the given values of "Number of Hours" as class boundaries, leading to deduction of marks. Some also found it challenging to locate the mode on the graph and drew the frequency polygon incorrectly.

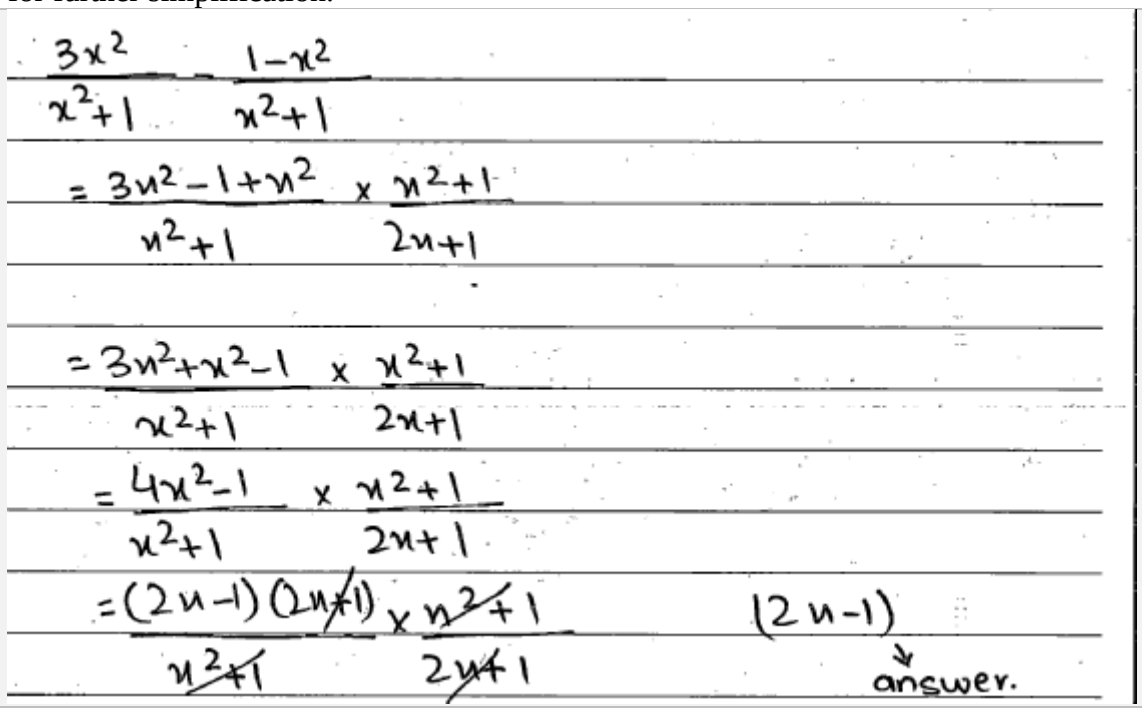
Image of Weaker Response	Estimate the mode of the data using graphical methods.  The mode of the given data is 12.

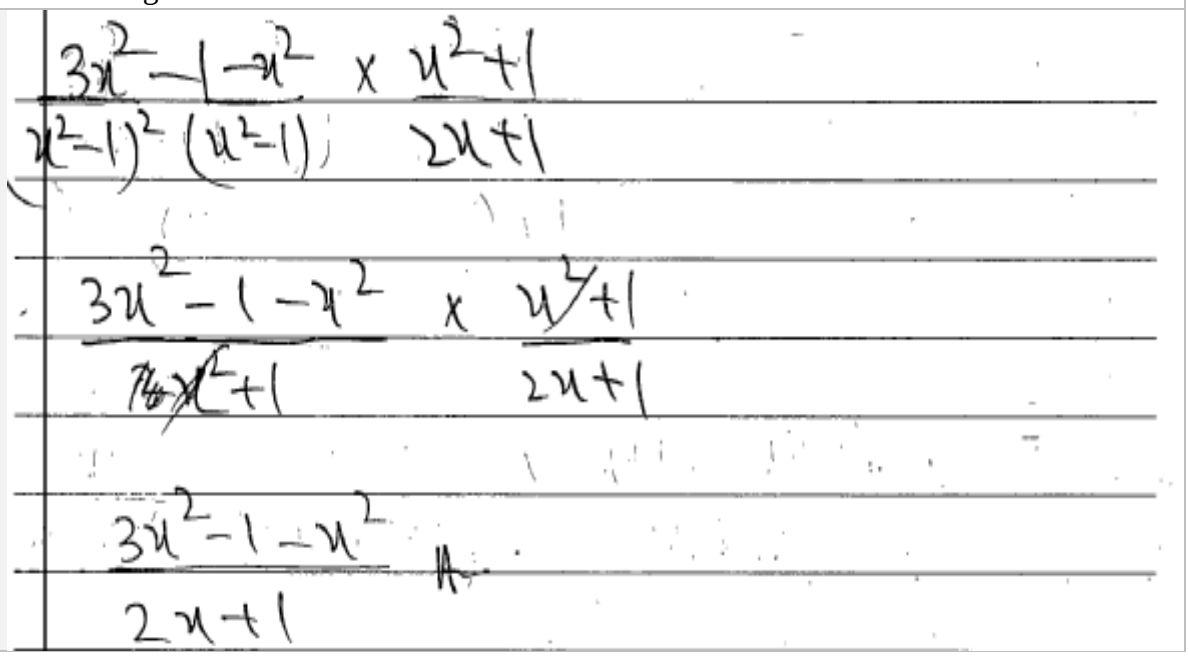
Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:		

Question No. 3a

Candidates were given the choice to attempt any ONE out of the two questions: 3a and 3b. Majority of the candidates attempted question 3a.

Question Text	Write the following algebraic expression in its simplest form. $\left(\frac{3x^2}{x^2+1}-\frac{1-x^2}{x^2+1}\right)\div\left(\frac{2x+1}{x^2+1}\right)$.
SLO No.	14.1.2
SLO Text	Simplify fractional expressions or rational expressions involving basic operations of ; +, -. *. /
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for taking L.C.M. 1 mark for simplification to get $\left(\frac{4x^2+1}{x^2+1}\right)$ 1 mark for getting $\left(\frac{(2x+1)(2x-1)}{x^2+1}\right)\times\left(\frac{x^2+1}{2x+1}\right)$ 1 mark to get the simplest form i.e. 2x - 1
Overall Performance	This question focused on the topic of simplification of algebraic expressions, requiring careful application and manipulation of algebraic techniques. Very few candidates were able to achieve full marks, with most candidates losing marks due to minor, yet avoidable, mistakes.
Description of Better Responses	Better responses carefully followed each step as required by the question. They simplified the terms enclosed in brackets by taking the LCM and applied straightforward identities for further simplification.
Image of Better Response	
Description of Weaker Responses	Weaker responses revealed that candidates struggled with simplification of algebraic expressions. They not only faced difficulty in finding the LCM of the denominators but also made errors in handling signs during numerator simplification. Additionally, they had difficulty applying identities for further simplification. These common mistakes resulted in

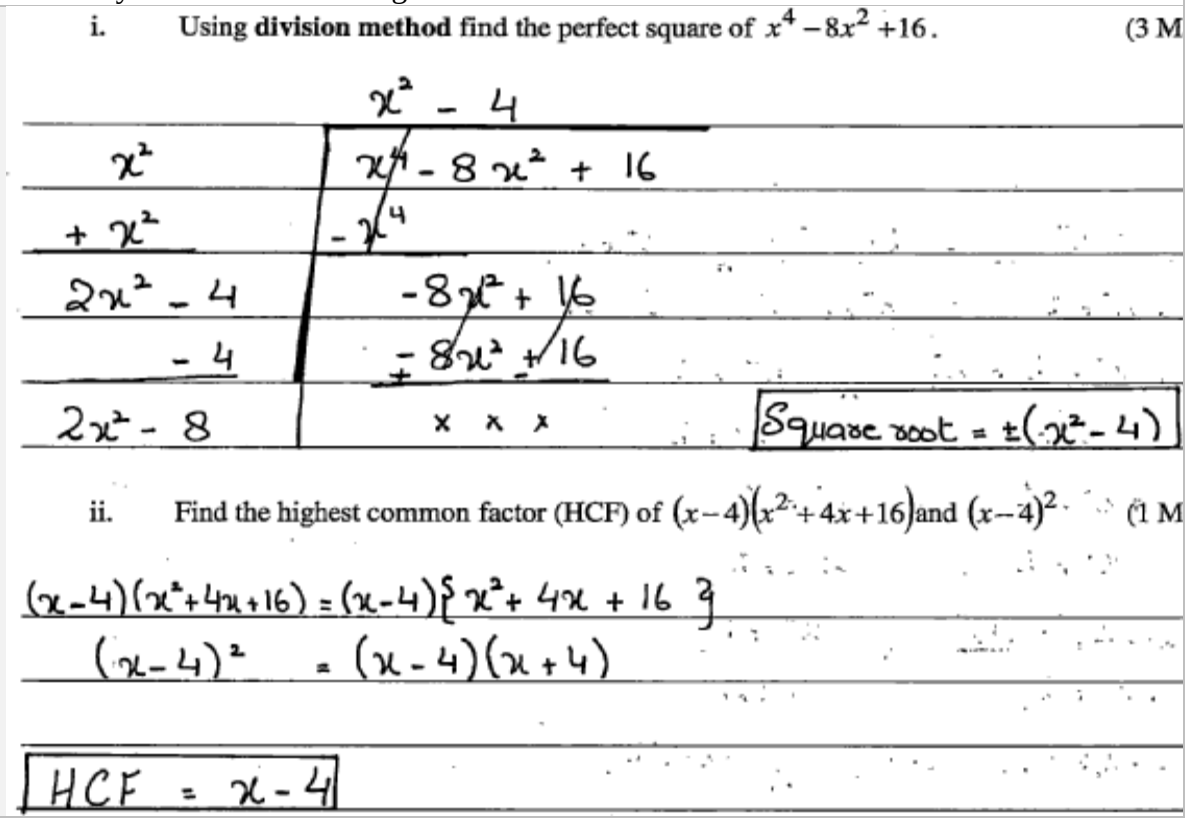
	them losing marks.
Image of Weaker Response	

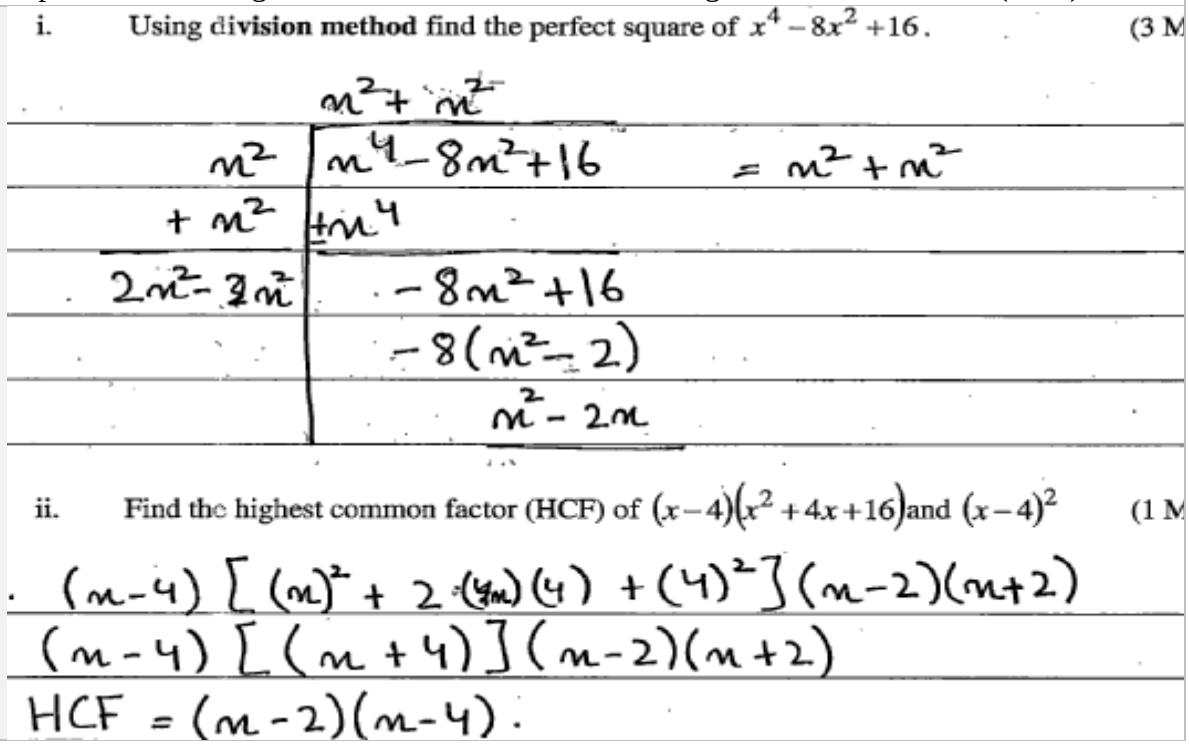
Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Pedagogy Used for that SLO**	**Assessment Strategies**
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Any Additional Suggestion:

1. Misconception Addressing: Be aware of common misconceptions, such as not multiplying negative sign inside the bracket, not changing the signs of all the terms inside that bracket e.g, -(1-x) is converting as -1-x, and address these issues in your teaching. Use examples to highlight the differences.
2. Error Analysis: Analyse common mistakes made by students when solving rational expressions. Encourage students to identify and correct their errors as part of the learning process.

Question No. 3b	
Question Text	i. Using **division method** find the perfect square of $x^4 - 8x^2 + 16$. ii. Find the highest common factor (HCF) of $(x-4)(x^2+4x+16)$ and $(x-4)^2$
SLO No.	14.2.1 and 14.1.2
SLO Text	Calculate the square root of algebraic expression by factorisation and division. Find the highest common factor (H.C.F) and the least common multiple (L.C.M) of algebraic expressions;
Max Marks	4
Cognitive Level	A
Checking Hints	i. 1 mark for each step (3 required) ii. Find the highest common factor (HCF) of $(x-4)(x^2+4x+16)$and $(x-4)^2$
Overall Performance	In this question, part i concerning square roots and the other related to finding the Highest Common Factor (HCF). Most candidates opt for option B, but only few manage to score full marks. A common mistake observed is the incorrect simplification of algebraic terms.
Description of Better Responses	Better responses indicated a clear understanding of both the division method for square roots and the factor method for finding the Highest Common Factor (HCF). The candidates carefully followed each step without making sign errors. In part (i), they correctly calculated and stated the square root of the given expression as required. In part (ii), they demonstrated understanding by accurately identifying factors and determining the necessary factor for calculating the HCF.
Image of Better Response	

Description of Weaker Responses	In part (i) of this question, many candidates correctly performed the initial step but made several mistakes during the simplification process and in identifying the second factor required to find the square root, resulting in deductions in their marks. In part (ii), candidates also struggled to identify the common factors in the given expressions, leading to incorrect calculations of the Highest Common Factor (HCF).
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestion:

Teachers may implement these teaching methodologies to develop a stronger foundation in this concept. Conceptual Clarity: Start with a strong emphasis on understanding the underlying concepts of HCF and its relationship to algebraic expressions. Ensure that students grasp the definition and significance of HCF. Step-by-Step Approach: Teach a systematic step-by-step approach to finding the HCF. Emphasise that the division method involves dividing both expressions and finding the common factors at each step.

Error Analysis: Encourage students to analyse their mistakes. When they make errors in adding or subtracting algebraic expressions, guide them in reviewing their work to identify where they went wrong.

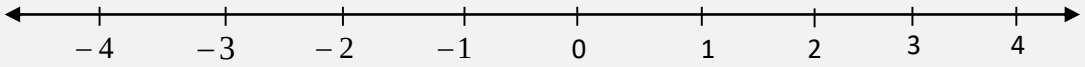
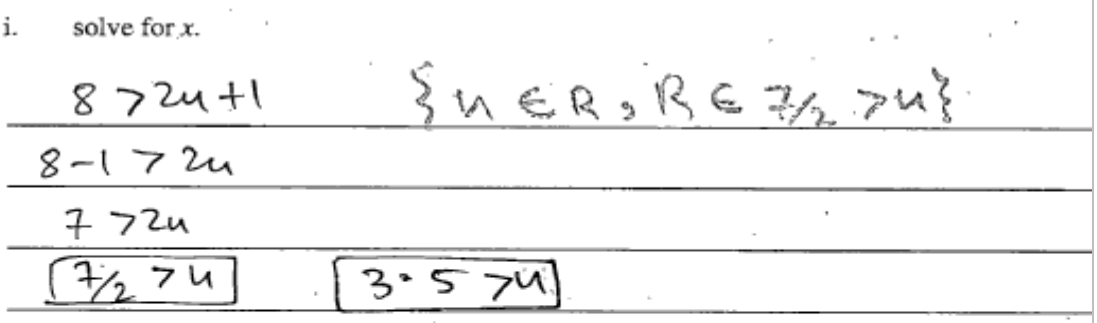
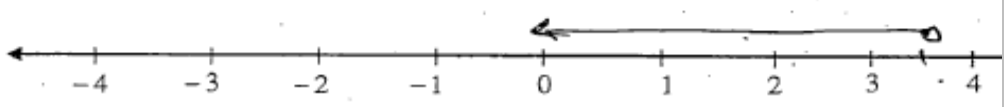
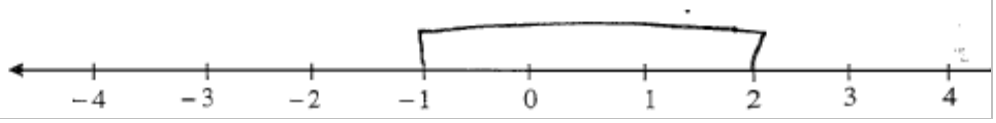
Question No. 4	
Question Text	If $x \in R$, then for $8 > 2x + 1$, i. solve for x. ii. show the solution on the given number line. 
SLO No.	15.4.1 & 15.4.2
SLO Text	• Solve linear inequalities with real coefficients in one variable. • Represent the solution of linear inequalities on the number line.
Max Marks	3
Cognitive Level	A
Checking Hints	i. 1 mark for subtracting 1 on both sides. 1 mark for dividing by 2 on both sides. ii. 1 mark for showing the solution on the number line.
Overall Performance	This question aims to assess the concept of inequalities in open sentences and their graphical representation. It was observed that many candidates struggled with handling inequalities in algebraic expressions, leading to errors in both their solutions and graphical representations.
Description of Better Responses	Better responses of this question demonstrated clear understanding of solving inequalities, accurately applying the signs of greater than and less than; and representing numbers by drawing the number line correctly.
Image of Better Response	i. solve for x.  ii. show the solution on the given number line. 
Description of Weaker Responses	Weaker responses revealed several misconceptions among candidates. Some made mistakes by incorrectly changing the sign of ‘greater than’ to ‘equals to’ and then solving accordingly. Others struggled to accurately represent the solution on the number line, indicating a lack of understanding of real numbers. These errors resulted in the deduction of marks.

Image of Weaker Response	If $x \in R$, then for $8 > 2x+1$, i. solve for x. $8 > 2x+1$ $2x = 8-1$ $x = 7/2$ ii. show the solution on the given number line. 

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

Following are some teaching strategies related to misconceptions and common errors which candidates exhibited.

Clear Definitions and Properties: Start by providing a clear and concise explanation of inequalities, including the greater-than (>), less-than (<), greater-than-or-equal-to (≥), and less-than-or-equal-to (≤) symbols.

Visual Aids and Number Lines: Utilise visual aids, particularly number lines, to help students visualise the solutions to inequalities. Demonstrate how to represent inequalities graphically and how to determine which side of the line represents the solution set.

Interactive Activities: Engage students in interactive activities where they can manipulate number lines and inequality symbols. Online tools and interactive whiteboards can be valuable for this purpose.

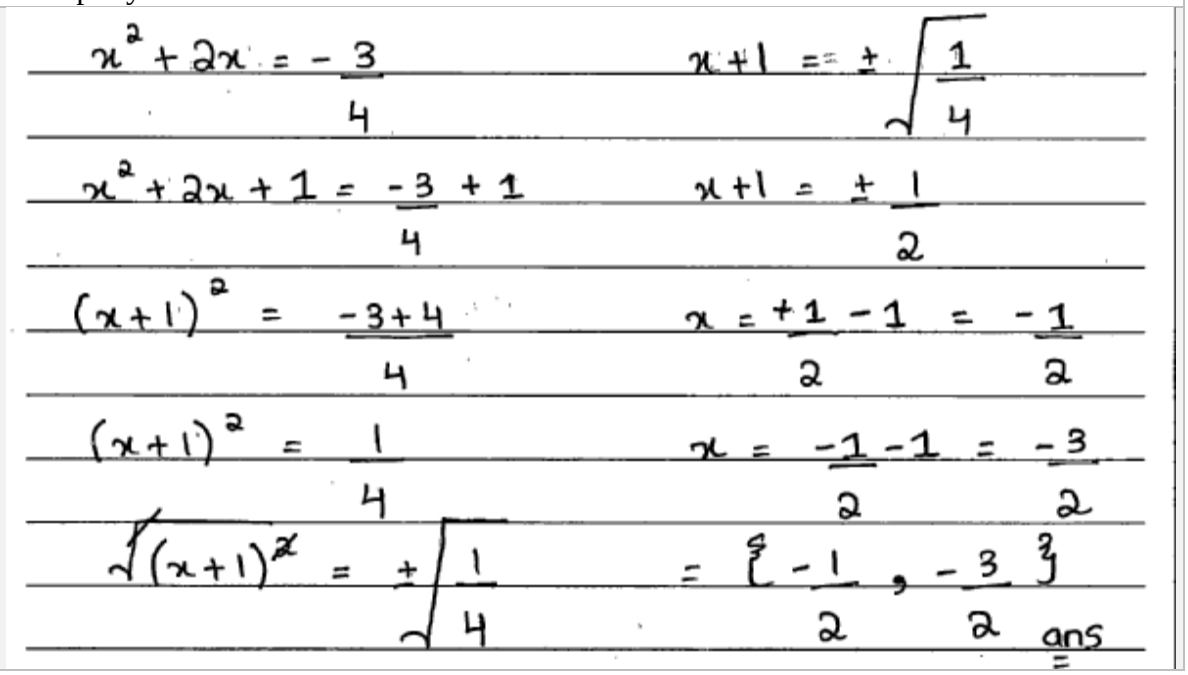
Real-Life Scenarios: Present real-life scenarios that involve inequalities, such as age restrictions, budget constraints, or temperature ranges. Encourage students to translate these scenarios into mathematical inequalities.

Step-by-Step Problem Solving: Teach students a systematic approach to solving inequalities, including the application of the multiplicative property of inequalities when multiplying or dividing by negative numbers (which requires reversing the inequality sign).

Error Analysis: Analyse common mistakes made by students, particularly those related to reversing inequality signs or the use of open and closed dots on number lines. Address these errors explicitly in your teaching.

Interval Notation: Introduce interval notation as an alternative way to represent solution sets. Show students how to express solutions using brackets [ ] and parentheses ( ), and explain the difference between open and closed intervals.

Technology Integration: Use technology to create interactive simulations and quizzes that allow students to practice solving inequalities and representing solutions graphically.

Question No. 5	
Question Text	Solve the equation $x^2 + 2x + \frac{3}{4} = 0$, by using completing square method.
SLO No.	16.2.1b
SLO Text	Solve a quadratic equation in one variable by b) completing square method.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for adding 1 to both the sides of the equation 1 mark for applying formula to write $(x+1)^2 = \frac{1}{4}$ 1 mark for taking square root to get $x+1 = \pm\frac{1}{2}$ 1 mark for finding the value of x in simplified form $x = -\frac{3}{2}$ or $x = -\frac{1}{2}$
Overall Performance	This question involved solving quadratic equations using the completing the square method, which demands skill in manipulating algebraic expressions. However, only few candidates demonstrated proficiency with this technique.
Description of Better Responses	Better responses showed that the candidates who grasped the concept of perfect square expressions, not only added the required term correctly on both sides but also carefully simplified each step of the question. One of such responses given below is considered exemplary.
Image of Better Response	

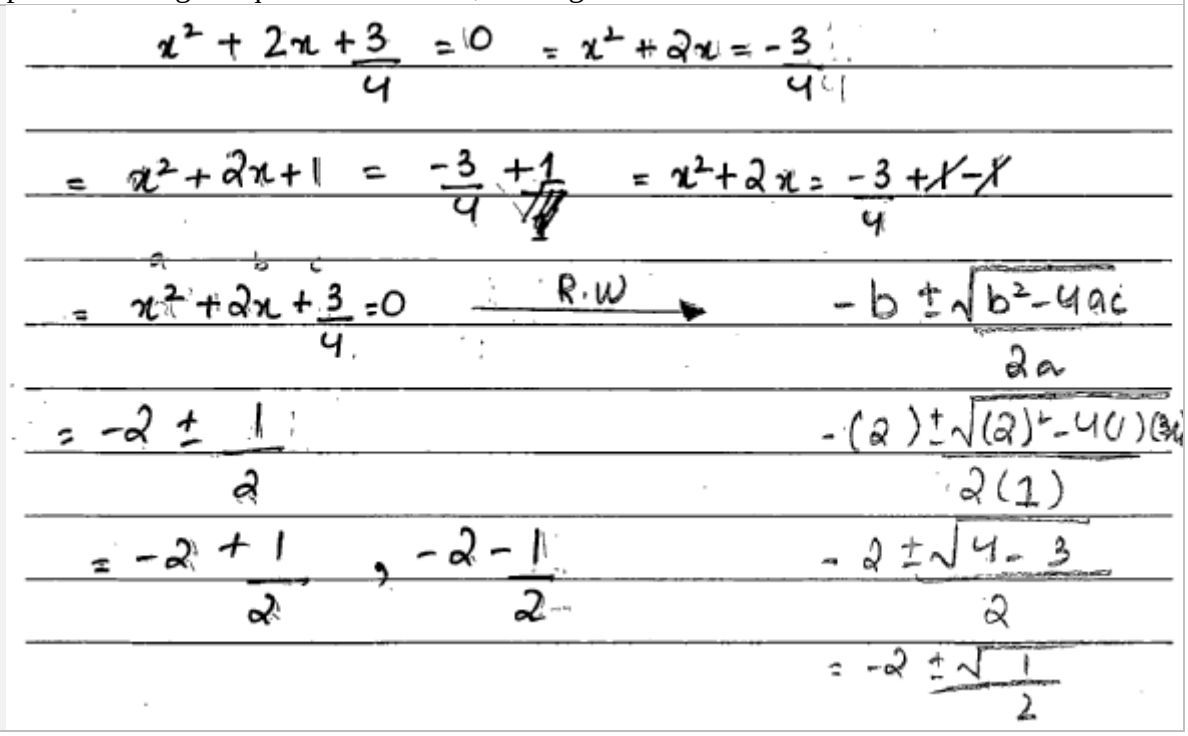
$x^2 + 2x = -\frac{3}{4}$ $\qquad$ $x+1 = \pm\sqrt{\frac{1}{4}}$

$x^2 + 2x + 1 = -\frac{3}{4} + 1$ $\qquad$ $x+1 = \pm\frac{1}{2}$

$(x+1)^2 = \frac{-3+4}{4}$ $\qquad$ $x = +\frac{1}{2} - 1 = -\frac{1}{2}$

$(x+1)^2 = \frac{1}{4}$ $\qquad$ $x = -\frac{1}{2} - 1 = -\frac{3}{2}$

$\sqrt{(x+1)^2} = \pm\sqrt{\frac{1}{4}}$ $\qquad$ $= \{-\frac{1}{2}, -\frac{3}{2}\}$ ans

Description of Weaker Responses	Weaker responses showed that candidates struggled with identifying the required term for adding to achieve a perfect square. In addition, candidates also made simplification mistakes such as while taking the LCM. Lastly, some candidates attempted to solve the problem using the quadratic formula, leading to deduction of marks.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

Teachers may implement these teaching methodologies to develop a stronger foundation in this concept. Concrete Examples and Visual Aids: Start with concrete examples where students can physically manipulate objects or tiles to form a square. Use visual aids like square tiles or diagrams to represent the process of completing the square.

Geometric Interpretation: Emphasise the geometric interpretation of completing the square. Show how the process relates to creating a perfect square trinomial, and how it transforms a quadratic equation into a square form.

Error Analysis: Present common mistakes that students make, such as adding or subtracting the wrong value when completing the square. Encourage students to analyse these errors and explain why they are incorrect.

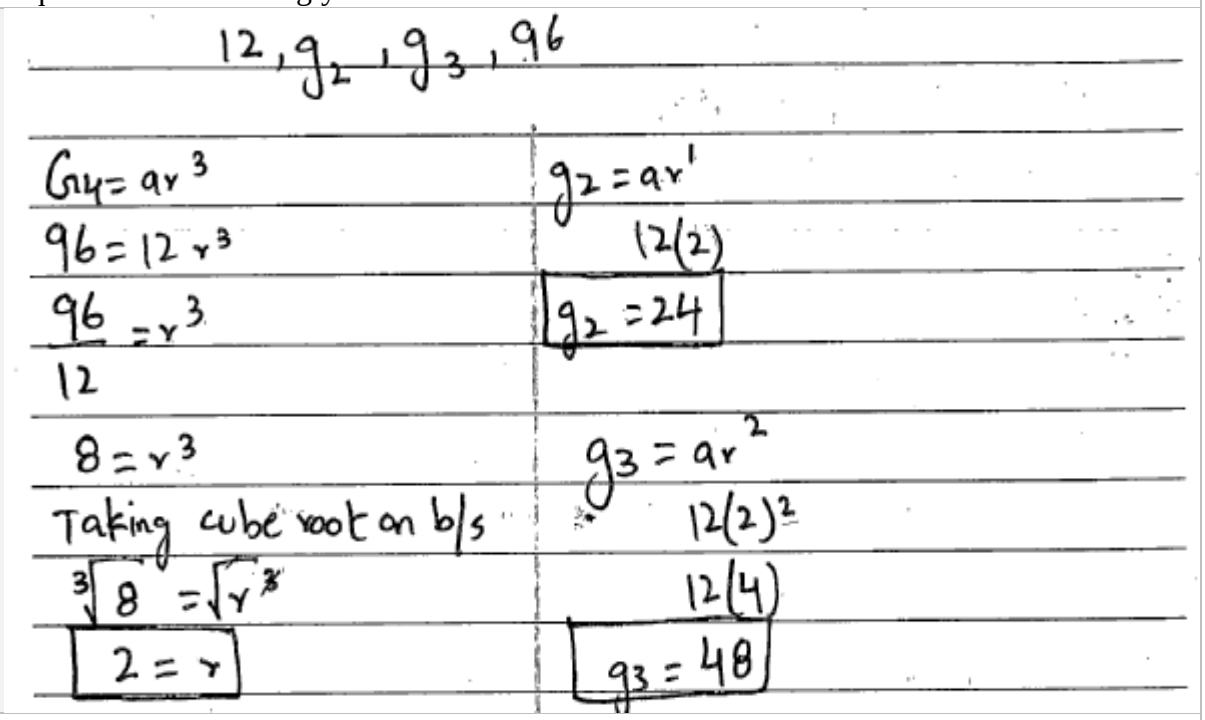
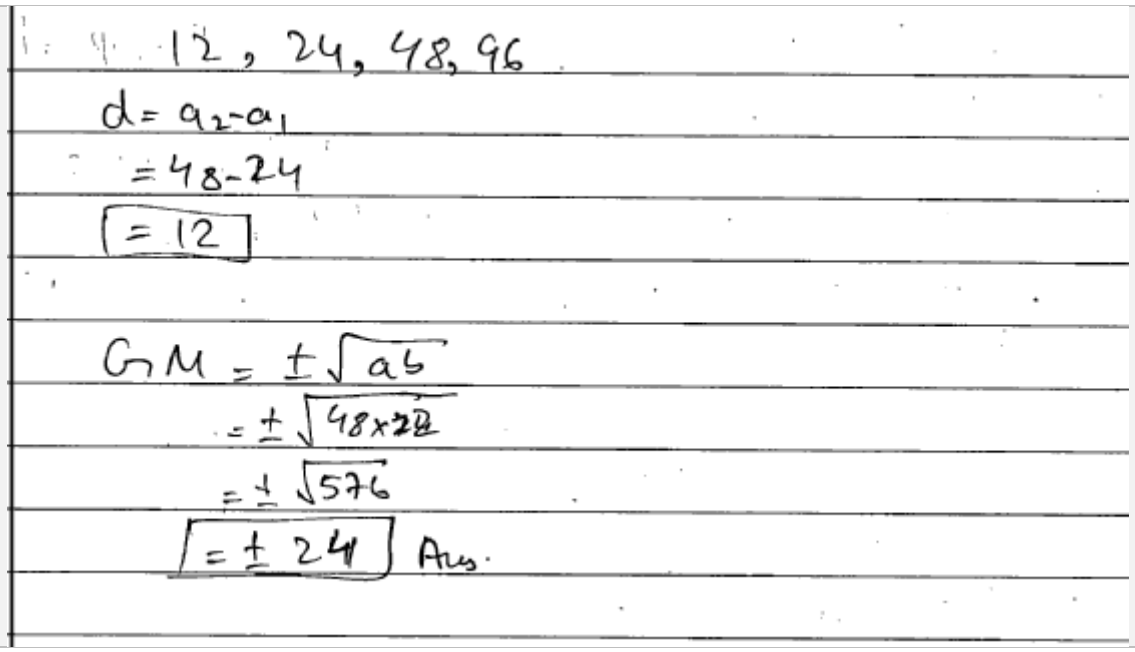
Question No. 6	
Question Text	Insert two positive geometric means between 12 and 96.
SLO No.	17.5.2
SLO Text	Find n G.Ms between two numbers, where n is a natural number.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for applying correct formula for calculating means 1 mark for finding r (or 2 marks if find *r* directly) 1 mark for each mean (2 required)
Overall Performance	The overall performance in this question indicates that many candidates did not fully grasp the question's requirements. Instead of applying the concept of geometric mean, they simply recognised the term 'mean' and attempted to calculate it. As a result, many candidates lost marks due to this misunderstanding.
Description of Better Responses	In this question, candidates who received full marks demonstrated a clear understanding of the requirements of the question. They correctly identified and constructed a geometric sequence. Moreover, they accurately determined the common ratio and calculated the required term accordingly.
Images of Better Responses	
Description of Weaker Responses	Many candidates lost marks because they did not clearly comprehend the question. Instead of using the formula for a geometric sequence as required, they attempted to calculate the geometric mean incorrectly, resulting in deduction of marks.

Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

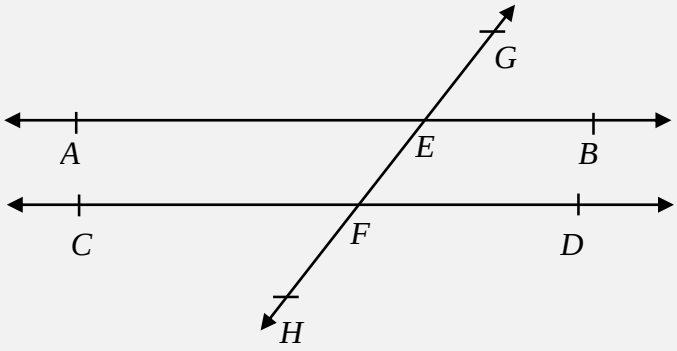
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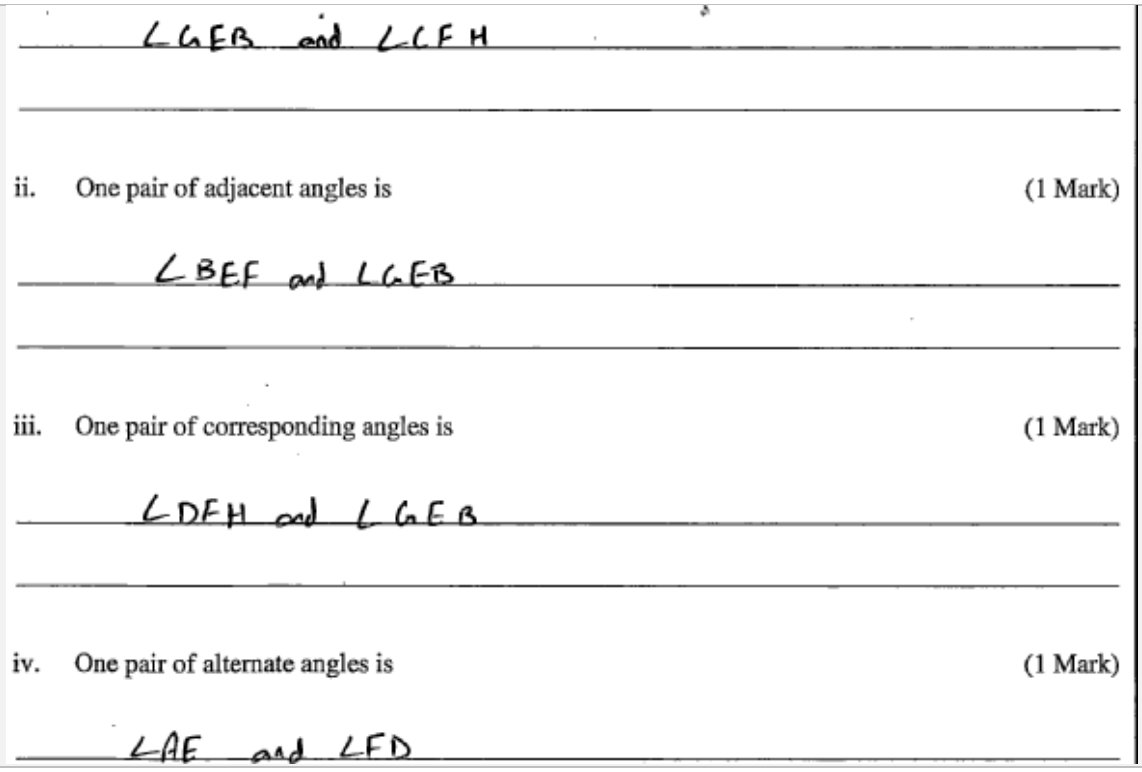
Use Visual Aids and Tools:

- **Diagrams and Graphs:** Use diagrams to visually illustrate the concept of geometric sequences and the process of inserting means. Graphs can help students understand how the terms in the sequence relate to each other.
- **Interactive Tools:** Incorporate digital tools or apps that allow students to explore and manipulate geometric sequences interactively.

Question No. 7a

Candidates were given the choice to attempt any ONE out of the two questions: 7a and 7b. Majority of the candidates attempted 7a.

Question Text	If in the given diagram, lines *AB* and *CD* are parallel to each other, then complete the following statements.  i. One pair of vertically opposite angles is ii. One pair of adjacent angles is iii. One pair of corresponding angles is iv. One pair of alternate angles is
SLO No.	18.2.2
SLO Text	Illustrate corresponding angles, alternate angles and interior angles.
Max Marks	4
Cognitive Level	U
Checking Hints	1 mark for writing for each correct answer (4 required)
Overall Performance	The responses of this question depict that Geometry served as challenging area for candidates specifically with reference to understanding and application of the key concepts such as vertically opposite angles, adjacent angles, corresponding angles and alternate angles. Though introduced in early grades, these concepts require rigorous reinforcement in classes.
Description of Better Responses	Better responses indicated that candidates not only correctly identified pairs of vertical, corresponding, adjacent and alternate angles but also specified their vertices. These candidates were awarded full marks for their clarity and accuracy.
Image of Better Response	i. One pair of vertically opposite angles is (1 Mark) $A\hat{E}H = G\hat{E}B$ ii. One pair of adjacent angles is (1 Mark) $C\hat{F}E = E\hat{F}D$ iii. One pair of corresponding angles is (1 Mark) $B\hat{E}F = H\hat{F}D$ iv. One pair of alternate angles is (1 Mark) $A\hat{E}F = E\hat{F}D$

Description of Weaker Responses	The responses to this question revealed that many candidates struggled not only to differentiate between alternate and corresponding angles but also confused vertical angles with adjacent angles. Despite being asked to select one pair out of four similar angles, they identified as many as they could find.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

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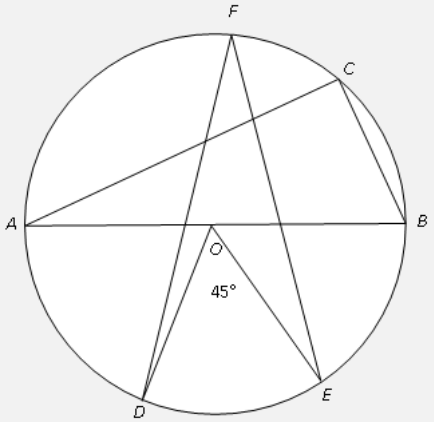
Any Additional Suggestion:

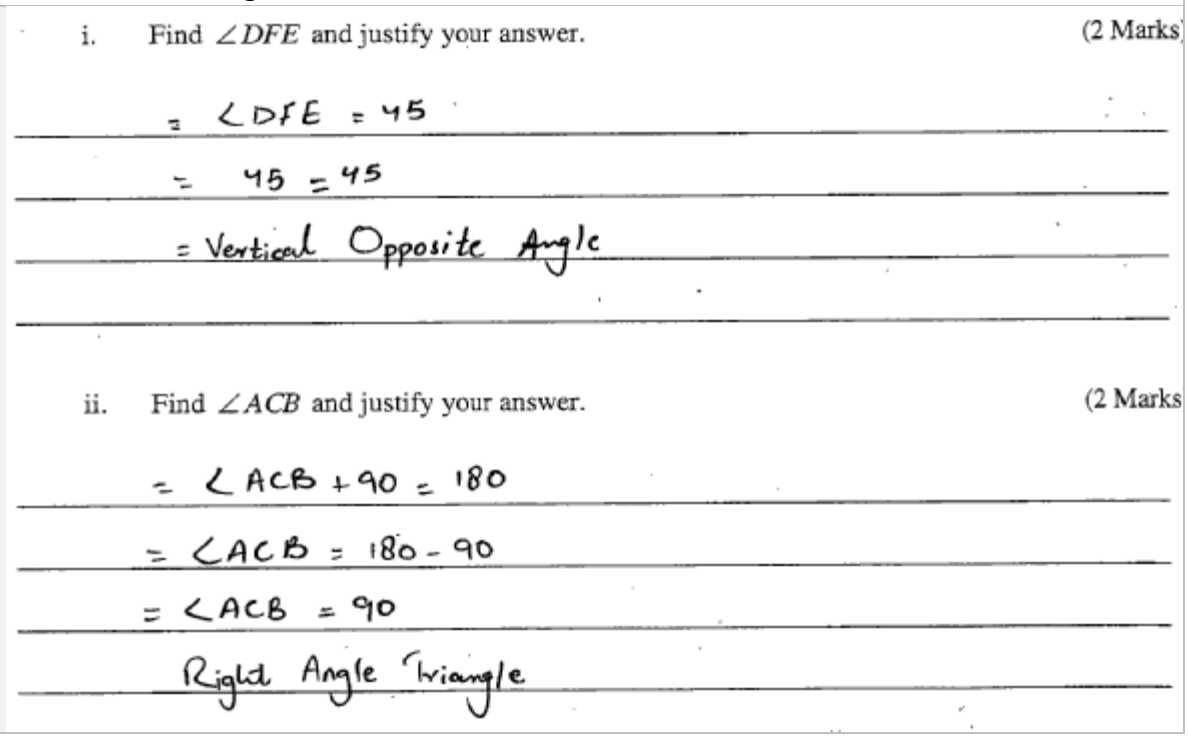
Strengthen Fundamental Concepts:

- **Review Definitions:** Ensure students understand the basic definitions and properties of corresponding, alternate, and interior angles.
- **Illustrate with Diagrams:** Use clear, labeled diagrams to show how these angles relate to parallel lines and transversals. Visual aids can help students grasp the concept better.

Question No. 7b

Candidates were given the choice to attempt any ONE out of the two questions: 7a and 7b.

Question Text	In the given figure, *O* is the centre of the circle and *AB* is the diametre.  i. Find $\angle DFE$ and justify your answer. ii. Find $\angle ACB$ and justify your answer.
SLO No.	18.5.2a, 18.5.2c
SLO Text	Apply the following properties in different geometrical figures: a. the angle in a semicircle is a right angle c. the central angle of a minor arc of a circle is double than that of the angle subtended by the corresponding major arc;
Max Marks	4
Cognitive Level	A
Checking Hints	i. 1 mark for finding $\angle DFE = 22.5^{o}$ 1 mark for writing reason to justify the answer ii. 1 mark for finding $\angle ACB$ 1 mark for writing reason to justify the answer
Overall Performance	Since this question is optional and pertains to circle properties, a small percentage of students attempted it. Very few achieved full marks, indicating that students struggled with applying theorems effectively in their responses.
Description of Better Responses	Better responses demonstrated both conceptual understanding and practical application of the theorem. They provided appropriate reasoning and accurately calculated the required angle, showcasing their knowledge of the relationship between inscribed angles and their corresponding central angles.
Image of Better Response	i. Find $\angle DFE$ and justify your answer. (2 Marks) DFE = $\frac{45}{2}$ = 22.5 ((central angle is twice the Opposite, and Opposite is half of Central angle) ii. Find $\angle ACB$ and justify your answer. (2 Marks) $\angle ACB = 90^{\circ}$ (Angle in a Semi Circle is 90°)

Description of Weaker Responses	Weaker responses struggled to identify the relationship between a central angle and its corresponding inscribed angle. Some incorrectly assumed triangle DFE to be isosceles and consequently calculated the wrong value. Others failed to provide the required reasoning, further contributing to their loss of marks.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

Teachers may implement these strategies to improve students' performance on this concept.

Clear Conceptual Foundation: Start by building a strong foundation in the basic properties of circles, such as radius, diametre, circumference, and area. Ensure that students have a clear understanding of these fundamental concepts.

Practical Applications: Discuss real-life applications of circle properties, such as calculating the area of circular objects, designing wheels, or understanding orbits in astronomy. This can make the topic more relatable and engaging.

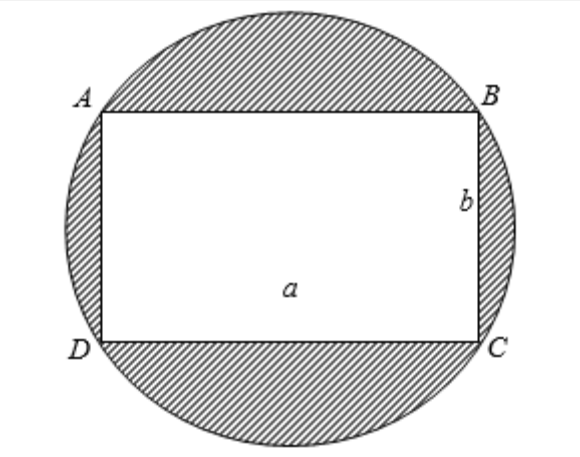
Question No. 8	
Question Text	In the given diagram, *ABCD* is a rectangle touching four points on the given circle internally.  If the radius of the circle is *r* cm, then i. show that the area of the shaded region is $(\pi r^2 - ab)$ cm^2. ii. calculate the area of the shaded region if a = 4 cm, b = 3 cm and r = 2.5 cm.
SLO No.	19.2.4
SLO Text	Solve problems related with areas of triangle, rectangle, square, parallelogram and circle.
Max Marks	4
Cognitive Level	A
Checking Hints	i. 1 mark for finding the area of the rectangle. 1 mark for finding the area of the circle. 1 mark for subtraction of the two areas. (or 3 if written directly) ii. 1 mark for arranging in terms of the required expression
Overall Performance	This question was about calculating the areas for which candidates need to use relevant formulas for the given shape. Overall, the performance has been average. Some candidates received full marks, while others lost marks due to not showing the proof of the area of the shaded region.
Description of Better Responses	Better responses were required to find the area of the shaded region in the given figure. Those with a strong understanding of calculating the area of composite shapes performed well. For part (i), candidates who approached the problem carefully and had a clear grasp of the formulas did well. In part (ii),candidates successfully calculated the numeric value by accurately substituting the values of a , b , and r , earning full marks.

Image of Better Response	i. show that the area of the shaded region is $(\pi r^2 - ab)$ cm^2. (3 Marks) area of circle $= \pi r^2$, $r = r$ $= \pi r^2$ area of rectangle $= l \times b$, $l = a$, $b = b$ $= ab$ area of shaded region $= \pi r^2 - ab$ cm^2 ii. calculate the area of the shaded region if $a = 4$ cm, $b = 3$ cm and $r = 2.5$ cm. (1 Mark) $\Rightarrow \pi(2.5)^2 - (4)(3)$ | cont: $\Rightarrow 19.625 - 12$ $\Rightarrow 3.14(6.25) - 12$ | $\Rightarrow$ 7.625 Ans. ~~19.625 - 12~~
Description of Weaker Responses	Weaker responses were unable to understand the question in part (i). Some of them were unable to find the area of rectangle in terms of a and b and some didn't do all the required steps to verify the given expression as the area of shaded region. Some candidates also made calculation errors while completing the solution in part (ii).
Image of Weaker Response	i. show that the area of the shaded region is $(\pi r^2 - ab)$ cm^2. (3 Marks) area of circle $= (1/2)^2 \pi r^2$ $(1/4)(3.14) r^2$ $0.785 r^2$ area $= 1/4 = \frac{C}{4} + \frac{R}{4}$ area $= \frac{C + R}{4}$ ii. calculate the area of the shaded region if $a = 4$ cm, $b = 3$ cm and $r = 2.5$ cm. (1 Mark) $(\pi r^2 - ab)$ cm^2 = Area $(3.14)(2.5)^2 - (4)(3)$ = area = 7.625

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

Teachers may implement these strategies to improve students' performance on this concept.

Clarify the Question: Train students to read and understand the question carefully. Ensure they grasp the specific requirement of the problem, such as calculating the shaded area.

Error Analysis: Share examples of common mistakes, such as using incorrect formulas or adding areas that should be subtracted. Ask students to identify and correct these errors.

Encourage Critical Thinking: Pose questions that require critical thinking, such as asking students to explain why they are adding or subtracting specific areas. Encourage them to justify their steps.

Visualise the Problem: Encourage students to visualise the problem by drawing diagrams. This can help them see the relationship between the given areas and the shaded portion.

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before

sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.

Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.
Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.
Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

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